

JINSEOK LEE

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Research Interests

Keywords: Soft Matter, Colloidal Science, Fluid Mechanics, Biophysics

I am a fluid mechanician by training, with a focus on understanding transport phenomena in deformable and crowded environments. My research aims to bridge conventional fluid mechanics with soft matter and biological physics, revealing fundamental mechanisms of transport in both synthetic and living systems.

Education

Yale University (Advisor: Dr. Yimin Luo)

Aug. 2024 – Current

Ph.D. in Mechanical Engineering and Materials Science,
Integrated Graduate Program in Physical and Engineering Biology (PEB)

United States

Sungkyunkwan University (Advisor: Dr. Jinkee Lee)

Feb. 2016 – Feb. 2024

B.S. and M.S. in Mechanical Engineering

South Korea

Thesis: “Investigation of cavitation bulk nanobubbles and its subsequent application in cyanobacterial inactivation”

Publications

Peer-reviewed publications

1. D. Pradhan, **J. Lee**, R. Prakash, A. Chauhan and J. Lee, “Tailored slip length: Investigating the effect of solid-liquid interfacial tension”, *Physics of Fluids*, 2025
2. **J. Lee***, R. Prakash*, S. Kim, H. Lee, and J. Lee, “Investigating the Influence of Cationic Surfactants with Varied Alkyl Chain Lengths on Nanobubble Characteristics”, *Journal of Molecular Liquids*, 2025, ***equally contributed**
3. R. Prakash*, **J. Lee***, S. Kim, H. Lee, and J. Lee, “An Insight into the Interaction between Ionic Nanobubbles and Cyanobacteria: Elucidating the Role of Surfactant and Zeta Potential in the Inactivation of Harmful Cyanobacteria”, *Chemical Engineering Science*, 2024, ***equally contributed**
4. R. Prakash, **J. Lee**, Y. Moon, D. Pradhan, S. Kim, H. Lee, and J. Lee, “Experimental Investigation of Cavitation Bulk Nanobubbles Characteristics: Effect of pH and Surface-Active Agents”, *Langmuir*, 2023, **Cover article**
5. D. Kim, **J. Lee**, A. Bose, I. Kim, and J. Lee, “The impact of an oil droplet on an oil layer on water”, *Journal of Fluid Mechanics*, 2021

Preprint & In preparation

1. T. Lin, **J. Lee**, M. Helgeson, M. T. Valentine, Y. Luo, and M. Gu, “Model-free estimation in scattering analysis of microscopy”, *In revision*
2. **J. Lee**, T. Lin, M. Gu, and Y. Luo, “Accessible pore geometry governs tracer diffusion in crowded environments”, *Under review*

Research Experiences

1. Flow-induced deformation in soft particulate porous media

Apr. 2026 – Present

Supervised by Dr. Yimin Luo (Yale)

- Investigating pressure-driven flow through packed beds of deformable PEGDA microparticles to elucidate the coupling between particle deformation, packing structure, and hydraulic resistance
- Performing controlled-flow experiments with time-resolved inlet–outlet pressure measurements to characterize transient pressure-drop evolution in soft particle packings
- Characterizing nonlinear pressure responses associated with flow-induced particle deformation, compaction, rearrangement, and changes in permeability
- Investigating mixed soft–rigid particle packings to determine how mechanical heterogeneity influences flow resistance, clogging, and structural stability

2. Confined diffusion of tracers in soft particulate liquids

Aug. 2024 – Present

Supervised by Dr. Yimin Luo (Yale); in collaboration with Dr. Mengyang Gu (UCSB)

- Revealed a transition from diffusive to confined tracer motion in crowded PEGDA microparticle matrices using passive microrheology experiments
- Developed a minimal Monte Carlo simulation that quantitatively reproduces experimental results

- Incorporated a data-driven predictive model to rapidly predict MSD curves from matrix geometric parameters (area fraction, particle size, and polydispersity)
- Demonstrated inference of environmental structure from tracer trajectories and validated the framework in intracellular transport systems

3. Eco-friendly nanobubble injector for green algae control Jun. 2021 – May. 2024

Supervised by Dr. Jinkee Lee (SKKU); in collaboration with Dr. Seung-Hyun Kim (Brown) and Dr. Ho-Yong Lee (SMU)

- Developed a cavitation-based nanobubble generator for freshwater algae mitigation
- Systematically tested environmental parameters (pH, surfactant type, input power) to optimize nanobubble properties
- Designed and fabricated a lead-free piezoelectric ultrasonic transducer inspired by pistol shrimp cavitation

4. Intravascular piezoelectric transducer for sonothrombolysis Jun. 2021 – Dec. 2023

Supervised by Dr. Jinkee Lee (SKKU); in collaboration with Dr. Ho-Yong Lee (SMU)

- Designed a minimally invasive intravascular ultrasound device for deep vein thrombosis treatment
- Improved thrombolysis efficiency using nanobubble-enhanced cavitation energy delivery

5. Oil droplet impact on immiscible layered liquids May 2018 – Jan. 2021

Supervised by Dr. Jinkee Lee (SKKU); with Dr. Ildoo Kim (Konkuk) and Dr. Arijit Bose (URI)

- Investigated the impact dynamics of oil droplets interacting with immiscible liquid layers
- Developed a dual-interface model describing droplet deformation, spreading, and interfacial coupling

Patents

1. “Green algae removal device and method”, KOR patent, *pending*, 2023,
2. “Nanobubble generating apparatus and method for controlling the size of nanobubble by using surfactant in acoustic cavitation method”, KOR patent, 10-2596334, 2023

Scholarships & Awards

Scholarships

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|---|---------------------|
| 1. Yale Graduate Fellowship, Yale University | Aug. 2024 – Current |
| 2. Future Technology Scholarship, Hyundai Motor Chung Mong-Koo Foundation | Sep. 2022 – Current |
| 3. Research Encouragement Scholarship, Brain Korea 21 SKKU | Sep. 2022 |
| 4. Graduate Merit Scholarship, SKKU | Feb. 2022 |
| 5. Jang Yeongsil Scholarship, SKKU | Mar. 2016 |

Awards

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| 1. IQB-MERCK Fellowship Awards, Institute of Quantum Biophysics, SKKU | 2023 |
| 2. 5th award, KSME-SEMES Open Innovation Challenge, KSME | 2022 |
| 3. Outstanding Paper Presentation Award, KSME Bioengineering Division Spring Conference | 2022 |
| 4. Encouragement Award, Creative Capstone Design Campus Competition, SKKU | 2019 |
| 5. 3rd prize, National Undergraduate Student Fluid Engineering Competition, KSME | 2018 |

Talks & Presentations

Invited Seminar Talk

- **Soft Fluid Living Matter Coffee Hour**, “Confined diffusion of tracers in soft particulate liquid”, Yale University, New Haven, CT, USA Jan. 2026

International Conferences

- **2026 IPoLS Annual Meeting**, “From Soft Particles to Living Cells: A Framework for Tracer Diffusion in Crowded Environments”, Munich, Germany Aug. 2026
- **100th ACS Colloids**, “Tracer Diffusion in Crowded Environments: From Soft Particulate Liquids to the Intracellular Environment”, Newark, DE, USA Jun. 2026
- **ASME-JSME-KSME Fluids Engineering Division**, “Mechanistic insight of bulk nanobubbles in various liquid mediums”, Osaka, Japan Jul. 2023

- **American Physical Society 75th Annual Meeting Division of Fluid Dynamics**, “Investigation of cavitation bulk nanobubbles in various pH and surfactant concentrations”, Indianapolis, IN, USA Nov. 2022

Workshops

- **102nd New England Complex Fluids**, “Non-Gaussian diffusion of tracers in soft particulate liquid”, Yale University, New Haven, CT, USA Mar. 2025

Korean Domestic Conferences

- **The Korean Society of Mechanical Engineers**, “Ionic Nanobubble-Mediated Inactivation of Harmful Cyanobacteria”, Incheon, South Korea Nov. 2023
- **KSME Micro/Nanoengineering Division Spring Conference**, “Effect of carbon chain length of the cationic surfactant on nanobubbles’ characteristics”, Busan, South Korea May 2023
- **The Korean Society of Visualization Fall conference**, “Analysis of the nanobubbles’ stability in various media”, Daejeon, South Korea Dec. 2022
- **The 12th National Congress of Fluids Engineering**, “Investigation of the effect of pH and surfactant on nanobubbles properties”, Changwon, South Korea Jun. 2022
- **KSME Micro/Nanoengineering Division Spring Conference**, “Analysis of the nanobubbles’ characteristics in various pH and surfactant solutions”, Busan, South Korea May 2022
- **KSME Bioengineering Division Spring Conference**, “Characterization of Cavitation Bulk Nanobubbles: Inspired by Pistol-shrimp”, Taejeon, South Korea Apr. 2022
- **The Korean Society of Mechanical Engineers**, “Analysis of Droplet Dynamics at Oil-Water Interface for Efficient Oil Dispersant Spraying in Oil Spill Accident”, Jeongseon, South Korea Dec. 2018

Teaching Experiences

Yale University

- **Mechanical Engineering I: Strength & Deformation (MENG 2311)** Sep. 2026 - Dec. 2026
Teaching Fellow
- **Mechanical Engineering III: Dynamics (MENG 3323)** Jan. 2026 - May. 2026
Teaching Fellow

Sungkyunkwan University

- **Design Lab on Thermo-Fluidics** Sep. 2023 - Dec. 2023
Teaching Assistant
- **Creative and Interdisciplinary Design** Feb. 2022 - Jun. 2022
Teaching Assistant

Skills

Programming: MATLAB (Image processing, Monte Carlo simulation), Python (Image processing)

Computational Tools: Ansys Fluent (CFD), SolidWorks (3D modeling), Inventor (3D modeling), Fiji(ImageJ) (Image processing), LaTeX(Academic writing) etc.

Military Service

Republic of Korea Army, Honorable Discharge

Army sergeant, Motor Transport Operator

Jul. 2019 - Jan. 2021

Army Aviation Operations Command